

Original Article

# Using Python Programming to Develop Quantity Takeoff Software with a Tkinter Graphical User Interface

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**Abstract** - This study aimed to develop a Quantity Takeoff software for calculating the quantity of material such as steel, concrete and formwork. Built using Python programming and a Tkinter graphical user interface. The application integrates traditional formula with automated tool for longitudinal bar weight calculation, stirrup quantity calculation, concrete volume calculation, formwork area calculation, and unit converter tool. For Validation using multiple case studies demonstrated zero deviation from conventional manual calculations and the accuracy of the software tool. The assessment workflow includes results from user trials, which revealed a 70% reduction in calculation time and a significant drop in human input errors from 2.3 down to just 0.2 errors per task. Application to a representative case study demonstrates significant impacts of functional integration, allowing users to seamlessly calculation many types of material in a time. Moreover, it can export all calculation data into excel.

**Keywords** - construction digitalization, material estimation, python programming, quantity takeoff, automated software.

## 1. Introduction

Quantity takeoff play an important role in construction industrial. Because it directly impacts on cost estimation of a project and budget planning. Recently, some engineer uses traditional methods such as spreadsheet excel, manual hand calculation, and web application.

Although Building Information Modeling (BIM) offers significant automation, but its adoption remains limited in the small-scale projects like small house due to high cost and computational requirements. Consequently, there is a critical need for a lightweight digital solution that bridges the gap between manual methods and BIM technology.

In this context, this study introduces a Quantity Takeoff software developed using python programming and a Tkinter graphical user interface. The objectives is to develop a quantity takeoff software for calculating the quantity of construction material such as steel, formwork, and concrete. It's can use for the small-scale projects. Also, this research using multiple case studies for validation from conventional manual calculations and the accuracy of the software tool.

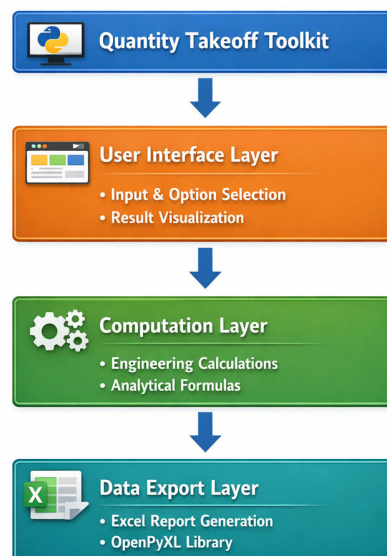
## 2. Materials and Methods

### 2.1. System Development Framework

The application was developed by using the python programming with Tkinter for the user interface. It's a lightweight desktop application that design as a standalone

and operates completely offline. To ensure modularity and functionality, the system is designed and divided into three main layers: user interface layer, computation layer, and data export layer using OpenPyXL.

The process will start working when the user inputs the data needed into the UI layer. By clicking button, variable is pushed into the computation layer, where the analytical formulations are processed and produce the result.



## 2.2. Reinforcement Weight Calculation

The formula for calculating the unit of rebar weight:

$$w = \frac{d^2}{162} \quad (1)$$

where:

$w$  = unit weight (kg/m)

$d$  = bar diameter (mm)

The total rebar weight is obtained as:

$$L_h = k \times d$$

where:

$n$  = number of bars

$L$  = bar lengths (m)

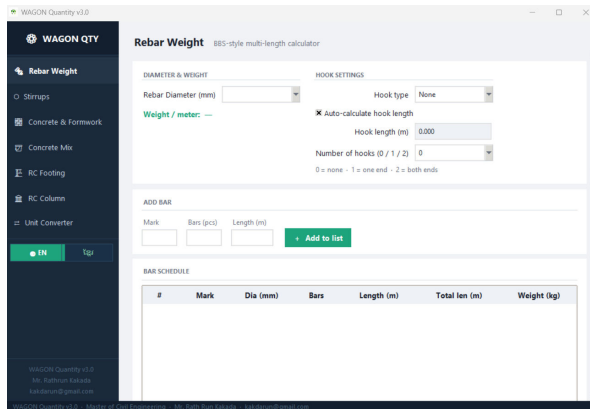
$L_h$  = hook length (m)

$n_h$  = number of hooks

For hook lengths are computed as a multiple of the bar diameter:

$$L_h = k \times d$$

For  $k$  depends on hook type (e.g., 10d, 12d, 16d).



$N$  = number of stirrups

$L$  = member length (m)

$s$  = spacing

### 2.3.2 Two-Zone Spacing

For structural members with different spacing zone:

$$N = 2 \left( \left\lceil \frac{L_e}{s_1} \right\rceil + 1 \right) + \left( \left\lceil \frac{L_m}{s_2} \right\rceil + 1 \right)$$

Where:

$L_e$  = end zone length

$L_m = L - 2L_e$

$s_1$  = end zone spacing

$s_2$  = midspan spacing

The stirrup length formula:

$$L_s = 2[(b - 2c) + (h - 2c)] + L_{hooks}$$

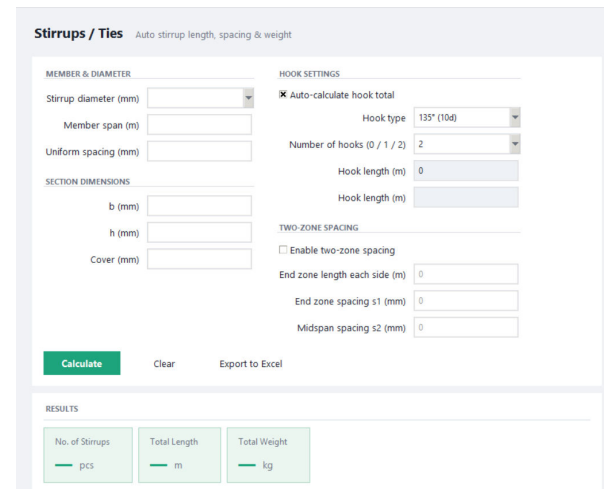
$L_s$  = length of stirrup

$L_{hook}$  = length of hook

$b$  = width of section

$h$  = height of section

$c$  = cover



## 2.3. Stirrup Quantity Calculation

### 2.3.1 Single-Zone Spacing

$$N = \left\lceil \frac{L}{s} \right\rceil + 1$$

Where:

## 2.4. Concrete Volume Estimation

Formula used to calculate the concrete volume:

$$\text{Beam: } V = b \times h \times L$$

$$\text{Slab: } V = L \times W \times t$$

$$\text{Rectangular Column: } V = b \times h \times H$$

$$\text{Circular Column: } V = \frac{\pi D^2}{4} H$$

Where:

$$V = \text{volume}(m^3)$$

$$L = \text{length of section}$$

$$W = \text{width of section}$$

$$h = \text{height of section}$$

$$t = \text{thickness}$$

$$H = \text{height of column}$$

$$D = \text{diameter}$$

### 2.5. Formwork Area Estimation

Formula for calculating the formwork area:

$$\text{Footing: } A = 2(b + L)H$$

$$\text{Beam: } A = (2h + b)L$$

$$\text{Column: } A = 2(b + h)H$$

$$\text{Wall: } A = 2 \times L \times H$$

$$\text{Slab: } A = 2(L + W) \times t$$

### 2.6. Validation Procedure

To verify the reliability and the operational efficiency of the application, a series of case studies representing real structure were prepared. The validation evaluates the proposed application against globally accepted the baseline methods. Each case was solved by using:

- 1- Manual hand calculation
- 2- Spreadsheet-based calculation
- 3- Developed application

The percentage error was determined as:

$$\text{Error} = \frac{|Q_{\text{tool}} - Q_{\text{manual}}|}{Q_{\text{manual}}} \times 100$$

Where  $Q_{\text{tool}}$  represents the material quantity by using the developed application, and  $Q_{\text{manual}}$  represents the material quantity by using traditional hand calculations.

## 3. Validation Using Case Studies

The validation process involves utilizing case studies from the user's direct experience with both methods. These

two approaches include the manual calculation method and the application method, aimed at obtaining accurate results (material quantities). The structural elements used for the quantity calculations include beams, columns, slabs, and reinforcement weight. The evaluation focuses on key output parameters such as total reinforcement weight, stirrup quantity, concrete volume, and formwork area. Moreover, the percentage error (E) was calculated to represented the level of agreement between both methods.

### 3.1. Reinforcement Weight Validation

This validation focuses on evaluating the reinforcement weight calculations for the selected structural column and beam cases. The parameters input is rebar diameter, hook settings, and rebar length. The results in table 1 obtained from the developed software and compared with manual hand calculations based on standard formulas. The objective is to verify whether the application can accurately generate reinforcement quantities under varying configurations.

**Table 1. Reinforcement Weight Validation**

| Case   | B    | N | L   | H      | M  | T  | E |
|--------|------|---|-----|--------|----|----|---|
| Beam   | 16mm | 4 | 6   | 2×135° | 37 | 37 | 0 |
| Column | 20mm | 8 | 3.5 | none   | 69 | 69 | 0 |

Note: B = bar diameter; N = number of bar (pcs); L = bar length (m); H = hook (deg.); M = manual(kg); T = toolkit(kg); E = error (%)

### 3.2. Stirrup Validation

The stirrup quantity calculation was evaluated for both uniform one-zone and two-zone spacing conditions. These cases studies were selected the structural beams to test. For beam No.1 with span 5.0m and uniform spacing of stirrup 150mm. For beam No.2 with span 5.0m and two zone spacing of stirrup 100mm at support and 200mm spacing at mid span. The comparison between hand calculations and application results aims to validate the correctness of the stirrup quantities and weight determination implemented in the software system.

**Table 2. Stirrup Validation**

| Case   | Span  | Spacing      | M  | T  | E |
|--------|-------|--------------|----|----|---|
| Beam 1 | 5.00m | 150mm        | 34 | 34 | 0 |
| Beam 2 | 5.00m | 100mm, 200mm | 40 | 40 | 0 |

Note: M = manual (pcs); T = toolkit (pcs); E = error (%)

### 3.3. Concrete Volume Validation

Concrete volume calculations also were validated using representative structural elements such as beams, slabs, and columns. The dimensions of each element were defined based on typical drawing. The result from the application used for compared with the hand calculations using standard formulas. For this validation ensures that the application accurately generates the correct volume of elements.

**Table 3. Concrete Volume Validation**

| Element | Dimension        | M    | T    | E |
|---------|------------------|------|------|---|
| Slab    | 4m x 3m x 0.15m  | 1.8  | 1.8  | 0 |
| Beam    | 0.3m x 0.5m x 6m | 0.9  | 0.9  | 0 |
| Column  | 0.4m x 0.4m x 3m | 0.48 | 0.48 | 0 |

Note: M = manual (m<sup>3</sup>); T = toolkit (m<sup>3</sup>); E = error (%)

### 3.4. Formwork Area Validation

Formwork area calculation was validated by comparing the hand calculated with the developed application. For evaluate the result, in this case studies will using slab, beam and columns for testing. After inputting the required data, the application generates the results of the formwork area of each element, and the results also were verified against the hand calculations. This comparison confirms that there is no error, and illustrate that it provides the same result as the hand calculations method.

**Table 4. Formwork Area Validation**

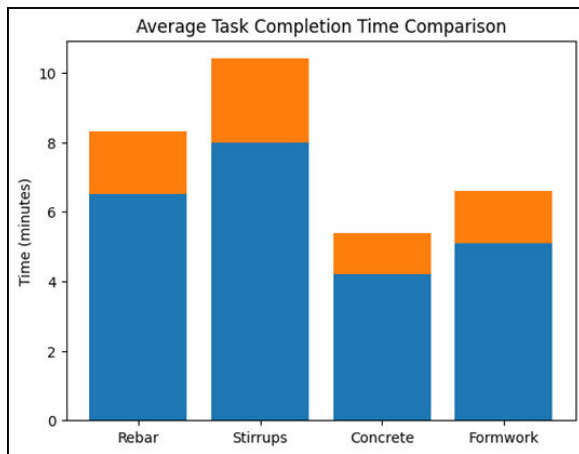
| Element | Dimension        | M    | T    | E |
|---------|------------------|------|------|---|
| Slab    | 4m x 3m x 0.15m  | 14.1 | 14.1 | 0 |
| Beam    | 0.3m x 0.5m x 6m | 7.8  | 7.8  | 0 |
| Column  | 0.4m x 0.4m x 3m | 4.8  | 4.8  | 0 |

Note: M = manual (m<sup>2</sup>); T = toolkit (m<sup>2</sup>); E = error (%)

## 4. Results and Discussion

### 4.1. Accuracy of Quantity Take off Calculations

The developed application called “Quantity take off software” was validated against manual hand calculations. Based on the case studies of structural elements, including slab, beam, column and the reinforcement weight. The comparison results (Tables in Section 3) indicate that the computed values from the application are the same to those obtained through manual hand procedures. The resulting in a percentage error of 0.00% for all tested cases.

**Fig. 1 Average Task Completion Time Comparison**

### 4.2. Time Efficiency Assessment

To evaluate workflow efficiency between manual hand calculation method and the developed application, the case studies of users performed identical quantity take off works was conducted.

**Table 5. Average Completion Time**

| Task                | M   | T   | Time Reduction (%) |
|---------------------|-----|-----|--------------------|
| Rebar calculation   | 6.5 | 1.8 | 72                 |
| Stirrup calculation | 8.0 | 2.4 | 70                 |
| Concrete volume     | 4.2 | 1.2 | 71                 |
| Formwork area       | 5.1 | 1.5 | 70                 |

Note: M = manual (minute); T = toolkit (minute)

The results shown on the table 5 is the average time reduction (%) of the completion time finished each task. The results illustrated that the user who used developed application can saved the time approximately 70%. For instance, the task of rebar calculation the user needs to spent probably 6.5 minutes to finish that task by using hand calculation method. So, for users who likely to perform multiple operations daily, this reduction translates into substantial productivity gains.

### 4.3. Error Reduction in User-Based Calculations

In addition to determining completion time, the user-generated errors were recorded during hand calculation and application workflows. Most manual hand calculation errors were associated with incorrect spacing conversion (mm → m), and missing to setting additional bar length for hooks. As detailed in table 6, the transition from hand calculation to developed application resulted in a profound improvement in computational accuracy.

**Table 6. Comparison of average user input errors**

| Method  | Average Input Errors per Task |
|---------|-------------------------------|
| Manual  | 2.3                           |
| Toolkit | 0.2                           |

### 4.4. Functional Integration and Workflow Improvement

A key advantage of the developed application is the integration of multiple functions into a single environment. In conventional practice, users typically switch between hand calculations, and spreadsheets. The proposed of the application is consolidates these operations into a unified workflow which performs the calculations, displays results immediately and exports all results reports to the excel file. This integration play an important role for reduces cognitive load and improves consistency in documentation.

### 4.5. Suitability for On-Site Engineering Applications

Unlike BIM-based quantity extraction, the developed application operates as a lightweight software, offline, and low-memory desktop application. This makes it particularly

suitable for site engineer works actually at the construction, small contractors, and also for the civil engineering students who working on the quantity take off assignments. Its tab-based interface allows users to transition quickly between function reinforcement, concrete, and formwork calculations without re-entering data into separate tools.

#### 4.6. Comparison with Existing Digital Approaches

While BIM platforms provide the quantity extraction directly from 3D models, but their implementation requires the Modelling works, and high software cost. In contrast, the proposed of developed application are requires only geometric input parameters such as dimension of section, amounts of rebar and select the function needed. After that produces immediate results and can export into excel file. Therefore, the system does not aim to replace BIM but to provide an intermediate digital solution between manual hand calculation and full model like Revit.

#### 4.7. Limitations of the Developed System

Despite its advantages, there are the several limitations that identified. The application requires manual input data such as geometric dimensions, length of rebar, and does not extract data from drawings automatically. For Hook length assumptions are based on predefined multipliers and may vary depending on design codes. Moreover, the system is not linked to the cost databases or scheduling modules. It is currently limited to complex shape of structural elements.

#### 4.7. User Survey Demographics

The majority of respondents for the user surveying were the site engineers and civil engineering students. And most participants having less than one year of professional experience. A significant proportion reported performing the quantity take off tasks on a daily works basis and also confirming the relevance of the survey sample.

##### - Respondent Role Distribution

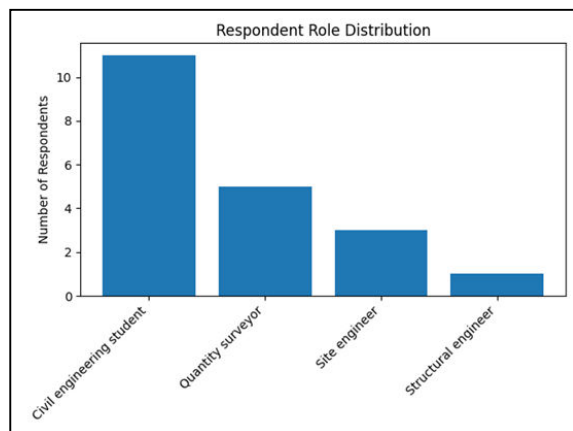


Figure. 2

##### - Years of Experience Distribution

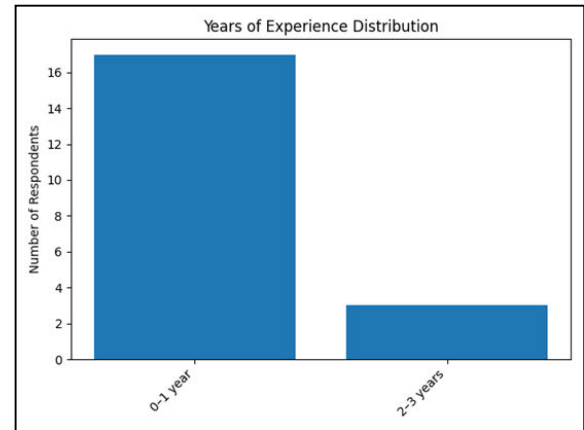


Figure. 3

##### - Quantity Take off Frequency

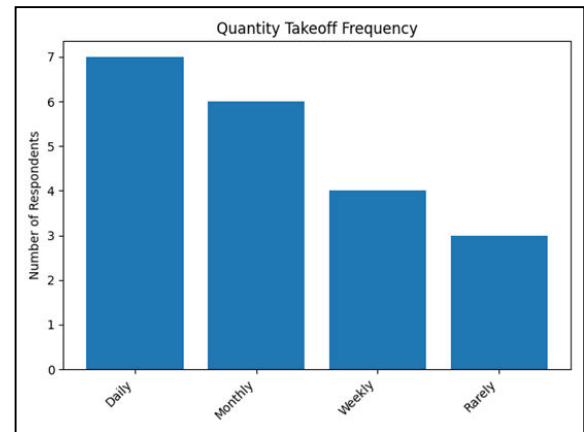


Figure. 4

##### - Perceived Efficiency and Accuracy

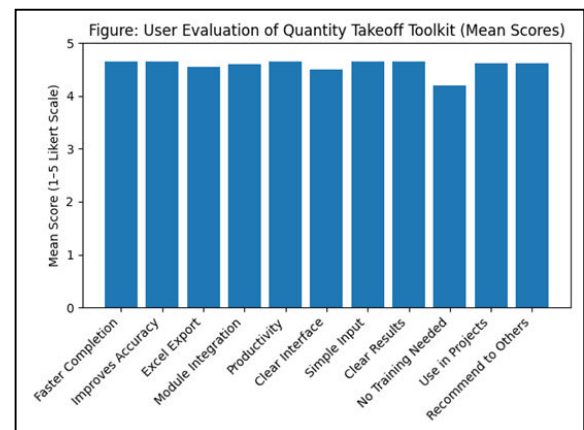


Figure. 5

According to the figure 1, 2, 3, 4, 5, the survey results illustrate that the most of the users was accepting of the developed application. Moreover, all the performance and usability were achieved high scores around 4.2 scores. Also, it has over 94% agreement on improving the productivity for the future update of application. The results confirm that the system is not only computationally accurate but also it is a user-friendly tool for real quantity work environments.

## 5. Conclusion

This study presented the software development for enhancing the quantity take off works and validation of a standalone software. The primary objective of this research was to provide a lightweight software with accurate and practical solution that bridges the gap between hand manual calculation methods and BIM-based systems. The developed application integrates multiple useful calculation modules including reinforcement weight calculation, stirrup quantity calculation with single zone and two-zone spacing, concrete volume calculation for structural elements, formwork area calculation, and the unit converter. In addition, after systems generated the results, user can export the results to the excel file. It's point to support standard documentation workflows at real site construction.

The findings result of validation clearly demonstrated that the outputs produced by the developed application are fully consistent with conventional hand calculations. For instance, with a percentage error of 0.00% for all tested of cases study. Particularly, this confirms the correctness of the embedded analytical models and the reliability of the computation engine of application. More importantly, the time-efficiency assessment revealed that the proposed system reduces the calculation time by approximately 70% compared with the manual hand methods. According to the validation results illustrate that the significant improvement point in productivity is attributed the system of application to the automatically operations by the user input standard structural drawing, built-in unit consistency, automatic hook length, and more advanced functionality.

While these findings provide valuable insights, this study had certain limitations. First, user-based evaluation further showed a substantial reduction in calculation errors.

For manual hand calculation were found to be prone to common mistakes such as inputting incorrect unit, excluded of hook lengths, and miscounting of reinforcement numbers. All these common issues were effectively minimized by the guided interface and automated procedures of the developed application. As a result, the application not only accelerates the process but also improves the overall reliability and standardization of quantity calculation.

Furthermore, the application is not yet integrated with cost estimation databases, the scheduling systems, or BIM technologies. Also, it's a standalone offline platform which is the users can only use in the desktop. There're no function or platform to use in the website or the mobile app. Nevertheless, despite these boundaries, the study offers a crucial foundation for the next renovation of this developed application. The future of the application development will focus on expanding the functionality of the application by incorporating:

- Automated bar bending schedule generation.
- Drawing-assisted input or model linkage.
- Cost estimation and material optimization.
- Mobile and web application.

In conclusion, the proposed developed application provides an accurate, fast and low-cost digital solution for the material quantity calculation. The system significantly improves computational efficiency, reduces human error and enhancing the workflow integration without the need for advanced modelling environments such as the BIM-technology. Additionally, this research demonstrates that small, dedicated engineering applications can have a major impact in digital transformation of construction practices. Particularly in resource-constrained environments, also it can enhance the low budget project with efficiency quantity works.

## Acknowledgement

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WAGON Quantity v3.0

**WAGON QTY**

Rebar Weight

**Stirrups**

Concrete & Formwork

Concrete Mix

RC Footing

RC Column

Unit Converter

EN

WAGON Quantity v3.0  
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### Stirrups / Ties

Auto stirrup length, spacing & weight

**MEMBER & DIAMETER**

Stirrup diameter (mm)

Member span (m)

Uniform spacing (mm)

**SECTION DIMENSIONS**

b (mm)

h (mm)

Cover (mm)

**HOOK SETTINGS**

☒ Auto-calculate hook total

Hook type

Number of hooks (0 / 1 / 2)

Hook length (m)

Hook length (m)

**TWO-ZONE SPACING**

☐ Enable two-zone spacing

End zone length each side (m)

End zone spacing s1 (mm)

Midspan spacing s2 (mm)

**Calculate** Clear Export to Excel

**RESULTS**

No. of Stirrups  pcs

Total Length  m

Total Weight  kg

WAGON Quantity v3.0

**WAGON QTY**

Rebar Weight

Stirrups

**Concrete & Formwork**

Concrete Mix

RC Footing

RC Column

Unit Converter

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### Concrete & Formwork

Quantity take-off by element type

**ELEMENT INPUTS**

Element type

Quantity (pcs)

L (mm):

W (mm):

t (mm):

**FORMWORK OPTIONS**

☐ Wall: both sides formwork

☐ Beam: 2 sides formwork

☐ Slab: include edge formwork

☐ Footing: include bottom formwork

☐ Column: 3 sides formwork

Concrete: —

Formwork: —

**Calculate** Clear

**BATCH ELEMENT LIST**

| Element type | Mark | Qty | Dim A (mm) | Dim B (mm) | Dim C (mm) | Option            |
|--------------|------|-----|------------|------------|------------|-------------------|
| Beam         |      | 1   |            |            |            | 3 sides (default) |

**+ Add**

| # | Mark | Type | Qty | Width (mm) | Length (mm) | Ht / Thk (mm) | Concrete (m <sup>3</sup> ) | Formwork (m <sup>2</sup> ) |
|---|------|------|-----|------------|-------------|---------------|----------------------------|----------------------------|
|---|------|------|-----|------------|-------------|---------------|----------------------------|----------------------------|



WAGON Quantity v3.0

### Concrete Mix Quantity

Material quantities per m<sup>3</sup> — M18, M20, M25, M30

MIX SELECTION & VOLUME

Concrete Grade:  Concrete Volume (m<sup>3</sup>):

Mix Ratio: 1 : 3 : 5 | Grade: M30 (1:3:5)

**Calculate** Clear Export to Excel

MATERIAL QUANTITIES

|                  |                  |                   |                  |                 |
|------------------|------------------|-------------------|------------------|-----------------|
| Cement           | Sand             | Gravel            | Water            | Admixture       |
| <b>388.00</b> kg | <b>752.00</b> kg | <b>1248.00</b> kg | <b>200.00</b> kg | <b>1.900</b> kg |

DETAILED BREAKDOWN BY UNIT

| Unit           | Cement | Sand   | Gravel  | Water  | Admixture |
|----------------|--------|--------|---------|--------|-----------|
| L              | 269.44 | 470.00 | 780.00  | 200.00 | 1.900     |
| kg             | 388.00 | 752.00 | 1248.00 | 200.00 | 1.900     |
| Bucket (20L)   | —      | 23.50  | 39.00   | 10.00  | 0.095     |
| Bag (50 kg)    | 7.76   | —      | —       | —      | —         |
| m <sup>3</sup> | 0.269  | 0.470  | 0.780   | 0.200  | 0.0019    |

Notes: L = Litres · Bucket = 20 L · Bag = 50 kg · m<sup>3</sup> = volume of material

QUANTITY PER STRUCTURAL ELEMENT (one-click add)

Element type:  L (mm):  W (mm):  t (mm):  Qty (pcs):  **+ Add element**

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WAGON Quantity v3.0

### RC Footing Calculator

Rebar · Formwork · Concrete · Lean Concrete

**+ Add Footing Type** **× Remove Current** **Recalculate All** **Export All → Excel**

**F1** **Summary**

FOOTING PARAMETERS

Footing Mark:  Footing Thickness t (mm):

No. of Footings (pcs):  Lean Concrete Thickness (mm):

Footing Length Lx (mm):  Concrete Cover (mm):

Footing Width Ly (mm):

BOTTOM REBAR

| Diameter | Spacing X(mm) | Spacing Y(mm) | No.Bars X | No.Bars Y | Take X | Take Y | BarLen X(mm) | BarLen Y(mm) |
|----------|---------------|---------------|-----------|-----------|--------|--------|--------------|--------------|
| DB20     | 150           | 150           | 10        | 16        | 10     | 16     | 1830         | 2730         |

**Total Bottom: —**

MIDDLE REBAR

| Diameter | Take No. X | BarLen X(mm) | Take No. Y | BarLen Y(mm) |
|----------|------------|--------------|------------|--------------|
| DB12     | 4          | 2850         | 4          | 4650         |

**Total Middle: —**

TOP REBAR

| Diameter | Spacing X(mm) | Spacing Y(mm) | No.Bars X | No.Bars Y | Take X | Take Y | BarLen X(mm) | BarLen Y(mm) |
|----------|---------------|---------------|-----------|-----------|--------|--------|--------------|--------------|
| DB12     | 150           | 150           | 10        | 16        | 10     | 16     | 1638         | 2538         |

**Total Top: —**

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WAGON Quantity v3.0

**WAGON QTY**

- Rebar Weight
- Stirrups
- Concrete & Formwork
- Concrete Mix
- RC Footing**
- RC Column
- Unit Converter

EN

**RC Footing Calculator** Rebar · Formwork · Concrete · Lean Concrete

+ Add Footing Type Remove Current Recalculate All Export All → Excel

**F1 Summary**

**TOP REBAR**

| Diameter | Spacing X(mm) | Spacing Y(mm) | No.Bars X | No.Bars Y | Take X | Take Y | BarLen X(mm) | BarLen Y(mm) |
|----------|---------------|---------------|-----------|-----------|--------|--------|--------------|--------------|
| DB12     | 150           | 150           | 10        | 16        | 10     | 16     | 1638         | 2538         |

Total Top: —

**RESULTS**

Formwork Concrete Lean Concrete

m<sup>2</sup> m<sup>3</sup> m<sup>3</sup>

**REBAR WEIGHT SUMMARY (kg)**

|          |   |    |          |   |    |
|----------|---|----|----------|---|----|
| RB6mm =  | — | kg | DB22mm = | — | kg |
| RB8mm =  | — | kg | DB25mm = | — | kg |
| DB10mm = | — | kg | DB28mm = | — | kg |
| DB12mm = | — | kg | DB30mm = | — | kg |
| DB14mm = | — | kg | DB32mm = | — | kg |
| DB16mm = | — | kg | DB36mm = | — | kg |
| DB18mm = | — | kg | DB40mm = | — | kg |
| DB20mm = | — | kg |          |   |    |

Total Rebar: — kg

Calculate Clear Export to Excel

WAGON Quantity v3.0  
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WAGON Quantity v3.0

**WAGON QTY**

- Rebar Weight
- Stirrups
- Concrete & Formwork
- Concrete Mix
- RC Footing
- RC Column**
- Unit Converter

EN

**RC Column Calculator** Rebar · Ties · Formwork · Concrete

+ Add Column Type Remove Current Recalculate All Export All → Excel

**C1 Summary**

**COLUMN PARAMETERS**

|                      |             |                         |     |
|----------------------|-------------|-------------------------|-----|
| Column Mark          | C1          | Width B (mm)            | 200 |
| Column Type          | Rectangular | Height H (mm)           | 200 |
| No. of Columns (pcs) | 2           | Diameter D (mm)         |     |
| Column Height (m)    | 4           | — B × H for rectangular |     |

**MAIN BARS**

| Diameter | No. of Bars | Bar Length (m)* | Total Length (m) | Weight (kg) |
|----------|-------------|-----------------|------------------|-------------|
| DB16     | 4           | —               | —                | —           |

\* includes lap splice (40d)

**TIES / STIRRUPS**

| Tie Diameter | Zone       | Spacing (mm) | No. Ties | Each len (m) | Total len (m) | Weight (kg) |
|--------------|------------|--------------|----------|--------------|---------------|-------------|
| RB8          | Beam zone  | —            | —        | —            | —             | —           |
|              | Mid column | —            | —        | —            | —             | —           |
|              | Slab zone  | —            | —        | —            | —             | —           |

Zone length each end (m) 0.8

Spacing @ Beam zone (mm) 100

Spacing @ Mid column (mm) 100

Spacing @ Slab zone (mm) 100

Total Ties: —

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**WAGON QTY**

- Rebar Weight
- Stirrups
- Concrete & Formwork
- Concrete Mix
- RC Footing
- RC Column**
- Unit Converter

EN

**RC Column Calculator** Rebar · Ties · Formwork · Concrete

+ Add Column Type Remove Current Recalculate All Export All → Excel

C1 Summary

Spacing @ Beam zone (mm) 100 Slab zone 100

Spacing @ Mid column (mm) 100

Spacing @ Slab zone (mm) 100

Total Ties: —

RESULTS

Formwork m<sup>2</sup> Concrete m<sup>3</sup>

REBAR WEIGHT SUMMARY (kg)

|             |             |
|-------------|-------------|
| RB6 = — kg  | DB22 = — kg |
| RB8 = — kg  | DB25 = — kg |
| DB10 = — kg | DB28 = — kg |
| DB12 = — kg | DB30 = — kg |
| DB14 = — kg | DB32 = — kg |
| DB16 = — kg | DB36 = — kg |
| DB18 = — kg | DB40 = — kg |
| DB20 = — kg |             |

Total Rebar: — kg

Calculate Clear Export to Excel

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WAGON Quantity v3.0

**WAGON QTY**

- Rebar Weight
- Stirrups
- Concrete & Formwork
- Concrete Mix
- RC Footing
- RC Column
- Unit Converter**

EN

**Unit Converter** Bidirectional, civil-engineering units

CATEGORY

Stress / Pressure

FROM

MPa

TO

psi

⇌ Swap

Type in either box — converts automatically

Clear

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**WAGON QTY**

Rebar Weight

Stirrups

Concrete & Formwork

Concrete Mix

RC Footing

RC Column

RC Beam

**RC Slab**

Unit Converter

EN

🌐

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RC Slab Calculator

Main Bar · Temp Bar · Concrete

+ Add Slab Type

✕ Remove Current

🔄 Recalculate All

📄 Export All → Excel

S1

Summary

SLAB PARAMETERS

Slab Mark

S1

No. of Slabs (pcs)

1

Slab Area (m<sup>2</sup>)

20

Slab Thickness t (mm)

150

Concrete Cover (mm)

20

MAIN BAR (SHORT DIRECTION)

Bar Diameter

DB12

Spacing (mm)

150

Span in direction (m)

4

(Short span - bars span this direction)

TEMPERATURE BAR (LONG DIRECTION)

Bar Diameter

DB10

Spacing (mm)

200

RESULTS

Concrete: —

Formwork: —

MAIN BAR RESULT

—

TEMPERATURE BAR RESULT

—

Concrete

Formwork

Main Bar

Temp Bar

Total Rebar

m<sup>2</sup>

m<sup>2</sup>

kg

kg

kg

REBAR WEIGHT SUMMARY (kg)

R86

R88

DB10

DB12

DB14

DB16

DB20

DB25

— — — — — — —

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